

What we've done so far

- Why is computing $f(x) \rightarrow f'(x)$ nice? its linear
- What should be true abt the domain and co-domain of a linear function? vector space!
- What is the minimal info needed to define $L: A \rightarrow B$?
it suffices to know the value of L on a basis for A
- What do $\mathbb{R}$ and $\mathbb{R}^2$ have in common? dimension $\rightarrow$ isomorphism "sameness for V space"
- how do vectors and linear functions look in coordinates?

Column of
Coefficients

matrix of columns
which represent the value
of the linear function on each
basis vector

Ex: linear func $F: \mathbb{R}^3 \rightarrow \mathbb{R}^2$

$\begin{matrix} & \nearrow \hat{i} & \nearrow \hat{j} & \nearrow \hat{k} \\ & & & \\ & & & \\ \nwarrow \hat{i} & \nwarrow \hat{j} & \nwarrow \hat{k} & \end{matrix}$

$$\begin{bmatrix} 3 & 4 & -2 \\ 1 & 0 & 0 \end{bmatrix}$$

$$F(1, 2, 3) = \begin{bmatrix} 3 & 4 & -2 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 3+8+(-6) \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix} = (5, 1)$$

Q: What do we know abt the composition of linear functions?

composition? $x \xrightarrow{f} x^2+1, x \xrightarrow{g} \ln x$

$$\mathbb{R} \xrightarrow{f} \mathbb{R} \xrightarrow{g} \mathbb{R}$$

$$\mathbb{R} \xrightarrow{f} (0, \infty) \xrightarrow{g} \mathbb{R}$$

$$(g \circ f)(x) = g(f(x)) = g(x^2+1) = \ln(x^2+1)$$

Lemma: The composition of 2 linear functions is linear

Proof is show that:

$$(L \circ K)(x+y) = (L \circ K)x + (L \circ K)y, \quad (L \circ K)(ax) = a(L \circ K)x$$

$$V \xrightarrow{K} W \xrightarrow{L} S$$

codomain of first func must equal the domain of the second func

$$\mathbb{R} \xrightarrow{K} \mathbb{P}_1 \xrightarrow{L} \mathbb{R}^2 \xrightarrow{L} \mathbb{R}^1 \quad \mathbb{R} \xrightarrow{K} \mathbb{P}_1 \xrightarrow{\Phi} \mathbb{R}^2 \xrightarrow{L} \mathbb{R}^1$$

must equal ($L \circ \Phi \circ K$)

What does composition look like in coords?

$$[1] \mathbb{R} \xrightarrow{K} \mathbb{R}^2 \xrightarrow{L} \mathbb{R} \quad K(1) = (1, 2) \quad L(1, 0) = 1 \quad L(0, 1) = 0$$

$\uparrow \downarrow$ $[1]$

What are the matrix reps of L and K $K: [-]$ $L: [- -]$

$$K: \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad L: \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$(L \circ K)(x) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} x \end{bmatrix} = x$$